

Program: Diploma in Fire Technology and Safety	
Course Code: 5461	Course Title: Safety in Process Industries and Power Handling
Semester: 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To understand ignition hazards and dust control in process industries.
- To explore electrostatic hazards and their control in powder handling.
- To learn safety measures related to power handling, particularly in hazardous environments involving electricity.
- To study best practices for handling industrial power and ensuring the safety of personnel.

Course Pre-requisites:

Topic	Course Code	Course Name	Semester
Principles of Safety Management		Principles of Safety Management	2
Electrical Safety		Electrical Safety	3
Environmental Safety		Environmental Safety	4

Course Outcomes

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand ignition hazards in process industries and dust control measures.	14	Understanding
CO2	Analyze electrostatic hazards, types of discharge, and control in powder handling.	15	Analyzing

CO3	Learn safety measures in power handling, particularly in high-risk situations..	15	Applying
CO4	Apply safety standards to prevent electrical accidents in power handling environments.	14	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	3					
CO3	3	3	2				
CO4	3	3	3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Understand ignition hazards in process industries and dust control measures.		
M1.01	Ignition hazards: minimum ignition energy, powder dispersion, and spark generation..	3	Understanding
M1.02	Characteristics of flammable gases, solvent vapor clouds, and exothermic and endothermic reactions.	4	Understanding
M1.03	Dust hazards: types, exposure, and dispersion; control strategies and monitoring techniques.	4	Understanding
M1.04	Control of dust at the source, occupational diseases, housekeeping, and environmental protection.	3	Applying

Contents:

Ignition hazards in powder handling and dust generation.
Characterization and control of flammable gases and vapor clouds.
Dust types and their control in industrial operations.
Occupational diseases and the importance of safety measures in process industries.

CO2	Analyze electrostatic hazards, types of discharge, and control in powder handling.		
M2 No	Description	Duration (Hours)	Cognitive Level
M2.01	Electrostatic charges in powder handling, energy release, and types of discharge.	4	Understanding
M2.02	Electrostatic discharge in powder handling operations and its impact.	3	Analyzing
M2.03	Health and safety risks associated with dust in industries such as chemicals, pesticides, and metal powders.	4	Analyzing
M2.04	Hazard identification and safety standards for industries handling dust and powder.	4	Applying
	Series Test I	1	

Contents:

Electrostatic hazards in powder handling and methods for mitigating these risks.
Health and safety risks due to exposure to fine powders in various industries.
Identification and control of hazards in powder processing industries.

CO3	Learn safety measures in power handling, particularly in high-risk situations.		
M3.01	Overview of electrical hazards and their impact on power handling.	3	Understanding
M3.02	Types of electrical faults and hazards in power systems.	4	Understanding
M3.03	Electrical shock prevention and safe handling of electrical equipment.	4	Applying
M3.04	Fire hazards due to electrical systems and fire safety protocols in power handling.	4	Applying

Contents: Safe Storage, Handling, and Environmental Management Overview of electrical hazards in industrial settings. Identifying and controlling electrical faults. Fire safety in power handling systems, including prevention of electrical fires.			
CO4	Apply safety standards to prevent electrical accidents in power handling environments		
M 4.01	Safe handling of power in high-risk environments, including isolation and grounding methods.	3	Applying
M 4.02	Emergency procedures and safety measures in case of electrical accidents.	4	Applying
M 4.03	Electrical protective systems: safety protocols and preventive measures.	4	Applying
M4.04	Safety audits and ensuring compliance with safety regulations in power handling.	3	Understanding
	Series Test II	1	
Contents: Electrical safety: Isolation, grounding, and preventive systems. Handling electrical emergencies, including power outages and faults. Safety audits in electrical and power handling operations			

Text /Reference:

T/R	Book Title/Author
R1	Crowl, D.A., Louvar, J.F., Chemical Process Safety – Fundamentals with Applications, Prentice-Hall, Pearson, 2011.
R2	Tate, L.P., Electrical Safety in Process Industries, Wiley, 2012.
R3	Warren, T.G., Safety in Process Industries, Elsevier, 2015.
R4	Cawley, J.G., Power System Protection and Safety, McGraw-Hill, 2016.
R5	American Institute of Chemical Engineers (AIChE), Guidelines for Engineering Design for Process Safety, AIChE, 2005.
R6	Tanner, M., Power Handling Safety and Industrial Regulations, Elsevier, 2014.